

WASP-36b

R-1991

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_150212 · created 2026-08-07T15:06:53+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457065.6937 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.154 +/- 0.048
Transit depth (Rp/Rs)^2	2.38 +/- 1.49 [%]
Orbital Inclination (inc)	80.74 +/- 2.96
Airmass coefficient 1 (a1)	0.99 +/- 0.53
Airmass coefficient 2 (a2)	0.03 +/- 0.15
Scatter in the residuals of the lightcurve fit is	52.0 %
Best Comparison Star	#1 - [189, 113]
Optimal Aperture	3.0
Optimal Annulus	3.5
Transit Duration (day)	0.049 +/- 0.032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.154 ± 0.048 vs 0.1368 (deviation 0.33σ)
Duration fit vs published	0.049 d vs 0.0758 d (deviation 0.77σ)
Mid-time O–C	-1.4 min
Depth SNR	2.9
Residual scatter	52.0 %

Light curve

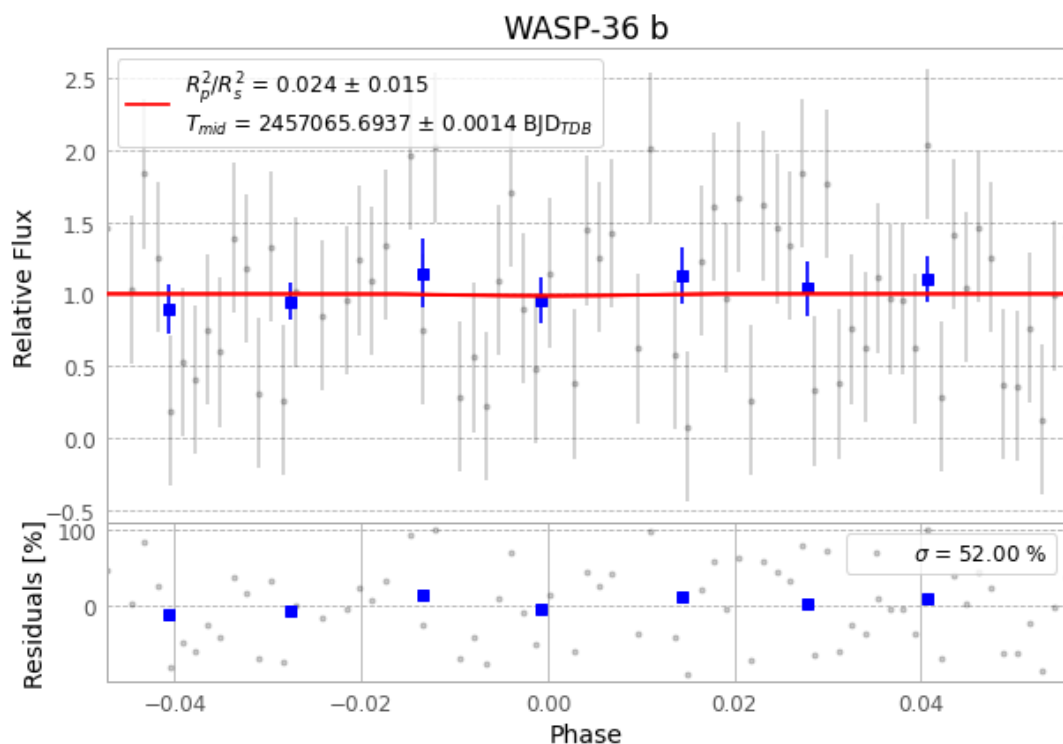


Figure 1 — FinalLightCurve_WASP-36 b_2015-02-11.png (SHA-256 6fe9f8d68aa60168...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [236, 367], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c22f26561aa8890b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

77 FITS frames (51 MB), WASP-36150212024512.FITS → WASP-36150212063312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-150212.log (d1cc939171ae...), FinalLightCurve_WASP-36 b_2015-02-11.png (6fe9f8d68aa6...), FinalLightCurve_WASP-36 b_2015-02-11.csv (04686a6975ef...)

Compute resources

Wall time 131.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 15:06Z.